Sample Research Document for SDG Testing

Title

AI-Based Smart Irrigation System for Sustainable Agriculture

Abstract

This research proposes an AI-based smart irrigation system using IoT sensors and machine learning to optimize water usage in agriculture. The system monitors soil moisture, temperature, humidity, and rainfall data to automate irrigation schedules and improve crop productivity while reducing water wastage.

Keywords

Artificial Intelligence, Smart Irrigation, IoT, Sustainable Agriculture, Water Management, SDGs

Introduction

Water scarcity is a major challenge affecting agriculture globally. This research focuses on developing a smart irrigation system capable of monitoring environmental conditions and making intelligent irrigation decisions.

Problem Statement

Traditional irrigation systems lead to excessive water consumption, low crop productivity, and high manual dependency.

Objectives

1. Develop an AI-based irrigation system.
2. Monitor soil moisture using IoT sensors.
3. Reduce water wastage.
4. Improve agricultural productivity.
5. Support Sustainable Development Goals.

Methodology

The system collects environmental data using sensors, processes the data using machine learning algorithms, and automatically controls irrigation motors.

Results and Discussion

The proposed system reduced water wastage by 35% and improved crop yield by 20% compared to traditional irrigation methods.

SDG Mapping

SDG 2: Zero Hunger
SDG 6: Clean Water and Sanitation

SDG 9: Industry, Innovation and Infrastructure
SDG 13: Climate Action

Conclusion

The AI-based smart irrigation system provides an effective and sustainable solution for modern agriculture.